Data Mining & Visualization (DMV)

Unit-Wise Viva Questions, Answers & Quick Revision Guide

B.E. AI & DS | Units 3 to 6 | SPPU 2020 Pattern

Unit 3 – Basics of Data Visualization

Part 1: Introduction to Data Visualization

Q1. What is Data Visualization?

Data visualization is the graphical representation of data using charts, graphs, maps, and plots. It helps us understand complex data easily.

Q2. Why is Data Visualization important?

- It makes raw data easy to understand and analyze.
- It helps quickly identify patterns, trends, and correlations.
- It assists in business decision-making.
- It highlights outliers and relationships between variables clearly.

Q3. What are the common types of Data Visualization?

Bar Chart, Line Chart, Pie Chart, Histogram, Scatter Plot, Box Plot, and Heatmap.

Q4. What is Computational Statistics?

Computational statistics uses computer algorithms, software, and statistical methods to analyze data, calculate metrics, and generate visualizations.

Q5. What is Exploratory Data Visualization?

It is the technique of using visual tools during data analysis to explore the dataset, spot patterns, discover relationships, and detect anomalies.

Q6. What is Presentation Graphics?

Presentation graphics are polished visualizations designed specifically to communicate final findings and analytical results clearly to a targeted audience.

Part 2: Graphs and Plots

Q7. What is a Bar Chart?

A bar chart represents categorical data using rectangular bars, where the height or length of each bar is proportional to the value it represents.

Q8. What is a Line Chart?

A line chart connects series of data points with line segments. It is primarily used to display continuous trends over time.

Q9. What is a Pie Chart?

A pie chart is a circular statistical graphic divided into slices, where each slice represents a numerical proportion of the whole dataset.

Q10. What is a Histogram?

A histogram displays the frequency distribution of continuous numerical data using adjacent vertical bars corresponding to data bins.

Q11. What is a Scatter Plot?

A scatter plot represents individual data points using dots on Cartesian axes to depict relationships or correlations between two numerical variables.

Q12. What is a Box Plot?

A box plot displays data distribution based on a five-number summary: minimum, first quartile (Q1), median, third quartile (Q3), and maximum, highlighting outliers.

Q13. What is a 3D Plot?

A 3D plot represents data across three spatial axes (X, Y, and Z) to display multi-variable relationships simultaneously.

Q14. What is High-Dimensional Display?

It refers to techniques used to visualize data with more than three dimensions using color encoding, markers, size, parallel coordinates, or dimensionality reduction.

Part 3: Graphics, Design & Iris Dataset

Q15. What are Statistical and Exploratory Graphics?

Statistical graphics visualize data distributions and metrics, while Exploratory graphics are initial plots used to investigate raw data structures before modeling.

Q16. What are good design choices in Data Visualization?

Choose appropriate chart types, use clear axes labels, provide descriptive titles, apply legible color contrast, and avoid clutter or deceptive scales.

Q17. What is the Iris Dataset?

The Iris dataset is a classic benchmarking dataset in machine learning containing 150 instances of iris flowers, 4 features (Sepal Length, Sepal Width, Petal Length, Petal Width), and 3 species classes (Setosa, Versicolor, Virginica).

Q18. Key Matplotlib commands in Python?

`plt.show()` (displays graph), `plt.xlabel()` / `plt.ylabel()` (labels axes), `plt.title()` (adds title), `plt.scatter()` (scatter plot), `plt.plot()` (line chart), and `plt.savefig()` (saves plot to image/PDF).

Last Minute Revision & Viva Test – Unit 3

Topic / Tool	Core Concept / Key Fact
Data Visualization	Graphical representation of data
Bar Chart vs Line Chart	Bar compares discrete categories; Line shows continuous trends over time
Histogram vs Box Plot	Histogram shows frequency distribution; Box plot shows quartiles and outliers
Iris Dataset	150 records, 4 features, 3 target species classes
Matplotlib / Pyplot	Python standard library and submodule for creating plots

1. **How many records are in the Iris dataset?** **Ans: 150**
2. **Which chart is best for trends over time?** **Ans: Line Chart**
3. **Which plot shows relationship between 2 numerical variables?** **Ans: Scatter Plot**

Unit 4 – Data Visualization and Data Wrangling

Part 1: Data Wrangling & Operations

Q1. What is Data Wrangling?

Data wrangling (or data munging) is the process of cleaning, transforming, and restructuring raw data into a usable format for analysis and visualization.

Q2. What is Hierarchical Indexing?

Hierarchical indexing (MultiIndex) allows working with multiple index levels on a single axis, enabling representation of higher-dimensional data in 2D structures.

Q3. What is the difference between Merge and Concatenate?

Operation	Key Difference	Pandas Function
Merge	Combines DataFrames based on common key columns (similar to SQL join operations).	`pd.merge()`
Concatenate	Stacks DataFrames together along a particular axis (rows or columns).	`pd.concat()`

Q4. What is Reshaping and Pivoting?

Reshaping reorganizes the structure of a DataFrame. Pivoting converts unique values from row entries into column headers to create summary views.

Part 2: Python Libraries & Graph Layout Techniques

Q5. Matplotlib vs. Seaborn?

Matplotlib is a low-level plotting library providing granular control. Seaborn is built on top of Matplotlib, offering high-level interfaces, integrated statistical functions, and attractive default themes.

Q6. What are Graph Representations, Nodes, and Edges?

Graph representation shows relational data. A **Node** (vertex) represents an entity, while an **Edge** represents the connection or relationship between two nodes.

Q7. What is a Force-Directed Graph and Bipartite Graph?

A **Force-Directed Graph** positions nodes using simulated physical forces (repulsion/attraction). A **Bipartite Graph** consists of nodes divided into two disjoint sets where edges only connect nodes across different sets.

Q8. What is Multidimensional Scaling (MDS)?

MDS is a visual dimension-reduction technique that projects pairwise similarities or distances of high-dimensional data points onto a lower-dimensional space.

Unit 5 – Data Aggregation and Analysis

Part 1: Aggregation & Split-Apply-Combine

Q1. What is Data Aggregation and Split-Apply-Combine?

Data aggregation gathers and summarizes dataset records. **Split-Apply-Combine** splits data into subsets based on keys, applies an aggregation function (sum, mean, count) to each subset, and combines results into a unified structure.

Q2. Difference between GroupBy and Pivot Table?

`groupby()` groups data by keys for element-wise calculations. `pivot_table()` summarizes data into a 2D grid matrix with customizable row/column headers and aggregation metrics.

Q3. Mean, Median, and Mode?

Mean is the arithmetic average (sensitive to outliers). **Median** is the middle value when sorted (robust to outliers). **Mode** is the most frequently occurring value.

Part 2: Time Series Analysis

Q4. What is Time Series Data and Resampling?

Time series data contains sequential observations recorded over time intervals. **Resampling** changes the time frequency of observations (e.g., aggregating daily data into monthly sums).

Q5. What is Moving Average?

A moving average calculates mean values across consecutive rolling time windows (using `df.rolling()`) to smooth out short-term fluctuations and highlight trends.

Important Commands Summary – Units 4 & 5

Command	Purpose / Utility
`pd.merge(df1, df2)`	Merges two DataFrames on shared key columns
`pd.concat([df1, df2])`	Concatenates DataFrames along specified axis
`df.pivot_table()`	Creates a multi-dimensional summary pivot table
`df.groupby('col').mean()`	Splits data by key column and computes mean
`df.resample('M').mean()`	Resamples time-series data to monthly frequency
`df.rolling(window=7).mean()`	Calculates a 7-period rolling moving average

Unit 6 – Data Analysis, Visualization and Modelling

Part 1: Medical Image Processing & Reconstruction

Q1. What is Medical Image Processing and Analysis?

Medical image processing involves techniques to enhance, segment, and analyze diagnostic medical images to assist healthcare professionals in identifying diseases and anatomical structures.

Q2. What is Medical Image Reconstruction?

Medical image reconstruction is the mathematical process of generating multi-dimensional anatomical images (2D/3D cross-sections) from raw sensor projections (such as CT or PET data scans).

Part 2: Modality Comparisons (PET, MRI, Ultrasound)

Imaging Modality	Physical Mechanism	Primary Clinical Application	Radiation Risk
PET (Positron Emission Tomography)	Detects gamma radiation emitted by injected radioactive tracers.	Visualizes metabolic activity, cellular function, and cancer detection.	Ionizing Radiation (Tracer)
MRI (Magnetic Resonance Imaging)	Utilizes strong magnetic fields and radio waves to align hydrogen atoms.	Provides high-detail structural imaging of soft tissues, brain, and joints.	No Radiation Risk
Ultrasound	Emits high-frequency sound waves and measures reflected acoustic echoes.	Real-time structural monitoring of organs, blood flow, and fetal development.	No Radiation Risk

Part 3: ER/Studio Case Study & Database Modeling

Q3. What is ER/Studio and its core application?

ER/Studio is an enterprise data architecture tool used to model, visualize, and design physical and logical database structures using Entity-Relationship diagrams.

Q4. Define Entity, Attribute, and Relationship in ER Diagrams.

- **Entity:** A real-world object or concept about which data is stored (e.g., *Student*, *Patient*).
- **Attribute:** A descriptive property or characteristic of an entity (e.g., *Age*, *Patient_ID*).
- **Relationship:** The logical association connecting two or more entities (e.g., *Patient assigned to Doctor*).

Comprehensive Viva Test – Units 5 & 6

1. **What are the three steps in GroupBy?** **Ans: Split, Apply, Combine**
2. **Which modality uses sound waves for imaging?** **Ans: Ultrasound**
3. **What type of information does a PET scan provide?** **Ans: Metabolic / Functional Activity**
4. **Which Pandas function handles rolling window calculations?** **Ans: rolling()**
5. **What does an ER Diagram model?** **Ans: Entities, Attributes, and Relationships**